Old Exam Questions
DL Architectures & Generative Techniques

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How this file is organized

Collected from the main exam (28.06.2024) and the 2022 / 2024 retry exams. Questions are grouped by topic so that the sections line up with the chapters of the course summary. Exact duplicates were removed; when essentially the same question recurred across exams it is marked with *Asked multiple times.* inside the box. Several originals were figure-based (drag-and-drop annotations, hand-drawn graphs, plots) — these are reproduced in words. The mathematical content and answer choices are kept intact; the answer key is at the end.

Contents

Image Classification & CNN Architectures

Question 1: AlexNet

Which of the following statements about AlexNet (Krizhevsky et al., 2012) are correct? (Select one or more.)

- ☐ AlexNet does not have fully-connected layers.
- ☐ AlexNet uses strided convolutions instead of max-pooling for down-sampling.
- ☐ AlexNet is a deep convolutional network with more than 1 hidden layer.
- ☐ One of the most important components of AlexNet is BatchNorm.
- ☐ AlexNet uses input images with a resolution of 224×224 pixels and three channels.

Question 2: VGGNet

Which of the following statements about VGGNet are correct? (Select one or more.)

- ☐ It predominantly uses 3×3 convolutions.
- ☐ It is a single-stage approach (a single 5×5 layer rather than stacked 3×3 layers).
- ☐ It has a symmetric encoder–decoder structure.
- ☐ The majority of the network's weights are located in the fully-connected layers.
- ☐ It cannot be used for image classification.

Question 3: Normalization in Image Classifiers

Which of the following statements about normalization in image-classification networks are correct? (Select one or more.)

- ☐ Batch Normalization has been used since AlexNet.
- ☐ Since Xception, most image classifiers started to use Layer Normalization.
- ☐ Batch Normalization was used in InceptionNet to speed up training.
- ☐ Batch Normalization is part of most deep networks for image classification (e.g. ResNet).

Question 4: Residual Networks

Which of the following statements about Residual Networks (ResNets) are correct? (Select one or more.)

- ☐ Residual networks use skip-connections that add the representation $\mathbf{h}^l$ of a former layer l to a later layer $l+1$, i.e. $\mathbf{h}^{l+1} = \mathbf{h}^l + F(\mathbf{h}^l)$, where F can be any transformation with appropriate dimensions (e.g. a fully-connected layer).
- ☐ Because of skip-connections, very deep ResNets can be trained even without normalization techniques.
- ☐ Pre-trained ResNets cannot be used for transfer learning because the learned features are usually overly specified to the task on which they were trained.
- ☐ Residual networks are always convolutional and cannot be used in purely fully-connected networks.

- ☐ Leaky ReLUs cannot be used in Residual Networks.

Question 5: 1×1 Convolution Output Size

You add a 1×1 convolutional layer with 16 kernels, padding 0, stride 1, to an input of shape $[128, 128, 64]$. Which statement is correct?

- ☐ It is not possible to apply this convolution layer.
- ☐ The resulting dimension is $[128, 128, 16]$.
- ☐ A 1×1 convolution does not exist.
- ☐ Better: replace the 1×1 layer with 2×2 max-pooling — an equivalent but more efficient operation.
- ☐ Better: use 1×1 with stride 2 — equivalent but more efficient.

Question 6: Receptive Field of the Last Layer

You are given a CNN with a fixed architecture. You may only change the parameters of the *last* convolutional layer, and you want to strongly increase the receptive field (RF) of each neuron in that last layer. For each change, decide whether it produces **no increase**, a **small increase**, or a **strong increase** of the RF.

- Increase the stride of the convolution ____________________
- Switch to dilated convolutions ____________________
- Increase the number of kernels ____________________
- Increase the kernel size ____________________

Question 7: Zero Initialization — Forward Pass

You design a standard 2-layer fully-connected network with sigmoid activations and SGD. You initialize all weights and biases to zero and forward-propagate an input $\mathbf{x}$. What is the output $\hat{y}$?

- ☐ 0
- ☐ e^x
- ☐ 0.5
- ☐ -1
- ☐ 1

Question 8: Zero Initialization — Weight Update

Consider the same all-zero-initialized network. You forward-propagate a batch, backpropagate, and update the parameters once. $\mathbf{W}^{(2)}$ denotes the weight matrix of the second layer. Which statement is then true?

- ☐ The entries of $\mathbf{W}^{(2)}$ are all positive.
- ☐ The entries of $\mathbf{W}^{(2)}$ may be positive or negative.
- ☐ The entries of $\mathbf{W}^{(2)}$ are all negative.

☐ The entries of $\mathbf{W}^{(2)}$ are all zero.

Semantic Segmentation

Question 9: Self-Driving Scene Labelling

A camera in a self-driving car provides road-scene images, and you train a network that outputs an image in which every pixel is annotated as car, road, pedestrian, etc. Which statements are true in this context? (Select one or more.)

☐ To train such a network, labels for each pixel of the training images are required.

☐ This represents an object-detection task in computer vision.

☐ Generative adversarial networks are the most appropriate type of network for this task.

☐ Deep networks cannot be used because there is no appropriate loss function.

☐ The intersection over union (IoU) would not be an appropriate metric to assess such a model.

Question 10: IoU Computation

A ground-truth object covers a set of pixels and a model predicts another set of pixels for that object, on a grid of pixels. The ground-truth region covers 10 pixels, the predicted region covers 8 pixels, and they overlap in 5 pixels. Compute the IoU metric, rounded to three decimals.

Answer: ________________

Object Detection & Localization

Question 11: Object-Detection Methods

Which of the following statements about deep-learning-based object-detection methods are correct? (Select one or more.)

☐ Faster R-CNN uses a region proposal network rather than the selective-search algorithm used in R-CNN.

☐ The operations RoI-pooling and RoI-align operate on feature maps and not on the input image.

☐ R-CNN uses AlexNet to extract features for Regions of Interest.

☐ Object detection can be formulated as a regression task in a multi-task setting.

☐ Fast R-CNN has two output layers (heads): one for regression and one for classification.

Question 12: One-Stage vs. Two-Stage Detectors

For each architecture, state whether it is a **one-stage** or **two-stage** detector.

- Faster R-CNN ____________________

- YOLO ____________
- Mask R-CNN ____________
- Fast R-CNN ____________
- R-CNN ____________
- SSD ____________

Question 13: Transpose Convolution (stride 1)

You are given an input $\mathbf{x} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$ and a kernel $\mathbf{w} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$. What is the result of the transpose convolution $\mathbf{w} *^T \mathbf{x}$ (stride 1)?

- ☐ (42)
- ☐ (14)
- ☐ $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \end{pmatrix}$
- ☐ $\begin{pmatrix} 13 & 13 & 13 \\ 13 & 13 & 13 \\ 13 & 13 & 13 \end{pmatrix}$
- ☐ $\begin{pmatrix} 0 & 2 & 1 \\ 6 & 14 & 6 \\ 2 & 3 & 0 \end{pmatrix}$
- ☐ $\begin{pmatrix} 3 & 2 \\ 1 & 0 \end{pmatrix}$
- ☐ $\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 2 & 0 & 3 \\ 4 & 6 & 6 & 9 \end{pmatrix}$
- ☐ $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 4 & 6 \\ 4 & 12 & 9 \end{pmatrix}$

Question 14: Transpose Convolution (stride 2)

You are given an input $\mathbf{x} = \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix}$ and a kernel $\mathbf{w} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$. What is the result of the transpose convolution $\mathbf{w} *^T \mathbf{x}$ with **stride 2**?

- ☐ $\begin{pmatrix} 0 & 1 & -1 \\ 2 & -1 & -1 \\ -4 & -2 & 6 \end{pmatrix}$
- ☐ $\begin{pmatrix} 3 & 2 \\ 1 & 0 \end{pmatrix}$
- ☐ $\begin{pmatrix} 0 & 1 & 0 & -2 \\ 2 & 3 & -4 & -6 \\ 0 & -1 & 0 & 2 \\ -2 & -3 & 4 & 6 \end{pmatrix}$
- ☐ $\begin{pmatrix} 0 & 2 & 1 \\ 6 & -14 & 6 \\ 2 & 3 & 0 \end{pmatrix}$

- ☐ (3)
- ☐ (42)

Autoencoders

Question 15: Annotating the Generative-Model Taxonomy

A figure shows three categories of generative models. Each row is a network with grey input/output boxes and a latent variable $\mathbf{z}$ in the middle. Annotate the building blocks with *Generator*, *Discriminator*, *Encoder*, *Decoder*, *Flow*, *Inverse Flow*, given that a rectangular block keeps the dimension constant while a trapezoidal block decreases or increases it.

- Top row: input $\to$ trapezoid $\to$ 0/1 $\to$ $\mathbf{z}$ $\to$ trapezoid $\to$ output

  ____
- Middle row: input $\to$ trapezoid (down) $\to$ $\mathbf{z}$ $\to$ trapezoid (up) $\to$ output

  ____
- Bottom row: input $\to$ rectangle $\to$ $\mathbf{z}$ $\to$ rectangle $\to$ output

  ____

Question 16: Autoencoders (excluding VAEs)

Which of the following statements about autoencoders (**not** including variational autoencoders) are correct? (Select one or more.)

- ☐ The input–output Jacobian of an autoencoder is always a lower-triangular matrix.
- ☐ Autoencoders do not map a single input to a single point in latent space, but to a distribution in latent space.
- ☐ Autoencoders are always fully-connected networks.
- ☐ All autoencoders use fewer neurons in the hidden layers than in the input layer.
- ☐ Autoencoders consist of an encoder and a decoder, which are parametrized neural networks.

Question 17: Sparse and Denoising Autoencoders

Which of the following statements about sparse / denoising autoencoders are correct? (Select one or more.)

- ☐ Sparse AEs enforce sparsity on the representation via a regularization term.
- ☐ Sparse AEs enforce hidden units to be inactive (i.e. activation $= 0$) for most inputs.
- ☐ A sparse AE can be trained on a copy task, where the input equals the target.
- ☐ Sparse AEs force the *input* units to be inactive (activation $= 0$).
- ☐ Sparse AEs enforce sparsity via early stopping.
- ☐ A sparse AE cannot be trained on a denoising task (input noisy, target clean).

Question 18: Autoencoder Reconstruction Loss

An autoencoder uses ReLU activations with encoder $u = \text{ReLU}\left(\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x}\right)$ and decoder $\hat{\mathbf{x}} = \text{ReLU}\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} u\right)$. Compute the reconstruction loss $\mathcal{L}(\mathbf{x}, \hat{\mathbf{x}}) = \sum_i (x_i - \hat{x}_i)^2$ for the input $\mathbf{x} = (1, 1, 2)^\mathsf{T}$.

Answer: ________________

Variational Autoencoders

Question 19: Variational Autoencoders

Which of the following statements about variational autoencoders (VAEs) are true? (Select one or more.)

- ☐ VAEs have an encoder–decoder structure, where encoder and decoder must be symmetric.
- ☐ The prior distribution of the latent variables $\mathbf{z}$ must be a discrete distribution.
- ☐ VAEs maximize a lower bound on the likelihood of the data.
- ☐ VAEs automatically provide the marginal likelihood $p(\mathbf{x})$ at low computational cost.
- ☐ VAEs cannot be used to generate images.

Question 20: VAE Objective Function

Using the notation from the lecture, which of the following is the correct formulation of the standard Variational Autoencoder (VAE) objective?

- ☐ $\mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(1 - D(G(\mathbf{z})))]$
- ☐ $\mathcal{L}_{\theta,\phi}(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}\,|\,\mathbf{x})} \log p_\theta(\mathbf{x}\,|\,\mathbf{z}) - D_{\text{KL}}(p_\theta(\mathbf{x}) \,\|\, p_\theta(\mathbf{z}))$
- ☐ $\mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[D(G(\mathbf{z}))]$
- ☐ $\mathcal{L}_{\theta,\phi}(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}\,|\,\mathbf{x})} \log p_\theta(\mathbf{x}\,|\,\mathbf{z}) - D_{\text{KL}}(q_\phi(\mathbf{z}\,|\,\mathbf{x}) \,\|\, p_\theta(\mathbf{z}))$
- ☐ $\mathcal{L}_\theta(\mathbf{x}) = D_{\text{KL}}(p_{\text{data}}(\mathbf{x}; \theta) \,\|\, p^*(\mathbf{x}))$

Question 21: Best Gaussian Approximation via KL

Asked multiple times.

You have a fixed, multi-dimensional, non-Gaussian distribution $p(\mathbf{x})$ that you want to approximate with a Gaussian $q(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, by minimizing $D_{\text{KL}}(p \,\|\, q)$ with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. Which expressions minimize the KL-divergence? ($\mathbb{E}$ is expectation, Cov the covariance, $\mathbf{I}$ the identity matrix.)

- ☐ $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x})$ and $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{x})$
- ☐ $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x})$ and $\boldsymbol{\Sigma} = \mathbf{I}$
- ☐ $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x}^\mathsf{T}\mathbf{x})$ and $\boldsymbol{\Sigma} = \mathbb{E}(\mathbf{x}^\mathsf{T})$
- ☐ $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x}^\mathsf{T}\mathbf{x})$ and $\boldsymbol{\Sigma} = \mathbf{I}$

☐ $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x})$ and $\boldsymbol{\Sigma} = \mathbb{E}(\mathbf{x}\mathbf{x}^\mathsf{T})$

GANs and Wasserstein GANs

Question 22: GAN Objective Function

Using the lecture notation, which of the following is the correct formulation of the standard objective of Generative Adversarial Networks (GANs)?

☐ $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(1 - D(\mathbf{z}))]$

☐ $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{x}\sim p_r}[\log(1 - D(G(\mathbf{x})))]$

☐ $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log G(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(1 - G(D(\mathbf{z})))]$

☐ $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(1 - D(G(\mathbf{z})))]$

☐ $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[D(G(\mathbf{z}))]$

☐ $\min_G \max_D \mathcal{L}(D, G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(D(G(\mathbf{z})))]$

Question 23: Properties of a Trained GAN

Consider a GAN that successfully produces images of apples. Which statements are correct? (Select one or more.)

☐ The discriminator can be used to classify images as apple vs. non-apple.

☐ The generator aims to learn the distribution of apple images.

☐ The generator can produce unseen images of apples.

☐ After training the GAN, the discriminator loss eventually reaches a constant value.

Question 24: Divergences and Distances

Which of the following statements about divergences and distances are correct? (Select one or more.)

☐ JS and KL divergences show problems for distributions with disjoint supports — the Wasserstein distance overcomes this.

☐ A map $D : \mathcal{S} \times \mathcal{S} \to \mathbb{R}$ is a divergence iff it is non-negative and symmetric.

☐ The Wasserstein-1 distance with Euclidean cost is a distance, not a divergence.

☐ The Wasserstein distance is defined as a minimization over joint distributions (couplings), but this is infeasible in practice.

☐ The Kantorovich–Rubinstein theorem lets us write the Wasserstein distance as a maximization over Lipschitz functions, which makes it tractable.

Question 25: KL, JSD and Wasserstein vs. Separation

Let $p_1 = \mathcal{N}(0, 1)$ and $p_2 = \mathcal{N}(x, 1)$. As x increases, you plot the KL-divergence $D_{\mathrm{KL}}(p_1 \,\|\, p_2)$, the Jensen–Shannon divergence $\mathrm{JSD}(p_1\|p_2)$, and the Wasserstein-2 distance $\mathrm{WD}(p_1\|p_2)$ against x. One curve grows quadratically without bound, one grows linearly, and one saturates at a finite value. Match each measure to its qualitative behaviour.

- KL-divergence ______________________
- Wasserstein-2 distance ______________________
- Jensen–Shannon divergence ______________________

Metrics, Normalizing Flows & Density Models

Question 26: Inception Score and FID

Tick the correct statements about the Inception Score (IS) and the Fréchet Inception Distance (FID). (Select one or more.)

- ☐ FID can detect mode collapse, because mode collapse influences the covariance matrix of the generated data.
- ☐ IS and FID are metrics that assess the quality of GAN-generated images.
- ☐ Both IS and FID rely on features of the input images produced by AlexNet.
- ☐ IS and FID can be used as GAN objectives that lead to faster training.
- ☐ The computation of the FID is based on an explicit formula for the Wasserstein-2 distance between two Gaussian distributions.

Question 27: Normalizing Flows

Which of the following statements about normalizing flows are correct? (Select one or more.)

- ☐ It is impossible to integrate a multi-layer perceptron as a transformation in a normalizing flow because MLPs are not invertible.
- ☐ All multi-layer perceptrons with the same input and output dimension can be considered a normalizing flow.
- ☐ Normalizing flows cannot contain functions with a lower-triangular Jacobian because they are not invertible.
- ☐ Some normalizing flows use functions with a lower-triangular Jacobian to facilitate the computation of the determinant.

Question 28: Flow-Based Models

Which of the following statements about flow-based models are correct? (Select one or more.)

- ☐ Theoretically, flow-based models can represent any distribution $p_x(\mathbf{x})$ if one starts from a sufficiently complex distribution; a simple base distribution $p_u(\mathbf{u})$ such as the uniform on the cube $(0,1)^D$ would not be sufficient.
- ☐ Fitting flow-based models is often done with the KL-divergence.
- ☐ Computing the Jacobi determinant of a neural network with D inputs and D outputs is in general of $\mathcal{O}(D^3)$.
- ☐ For autoregressive flows the transformation g_ϕ is constructed so that its Jacobi determinant always has value 1.

Question 29: Density under a Linear Transform

Asked multiple times.

Assume a two-dimensional vector $\mathbf{u} \sim p_u(\mathbf{u})$. Let $T = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$ and $\mathbf{x} = T \cdot \mathbf{u}$. Mark the correct formula for the distribution $p_x(\mathbf{x})$.

- ☐ $p_u\left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix} \mathbf{x}\right) \cdot \frac{1}{2}$
- ☐ $p_u\left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix} \mathbf{x}\right)$
- ☐ $p_u\left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x}\right) \cdot \frac{1}{2}$
- ☐ $p_u\left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x}\right)$
- ☐ $p_u\left(\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x}\right) \cdot 2$
- ☐ $p_u\left(\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x}\right) \cdot \frac{1}{2}$

Question 30: Density under a Linear Transform (Non-Diagonal)

Let $\mathbf{u} \sim p_u(\mathbf{u})$ be a two-dimensional random vector and define $\mathbf{x} = T\mathbf{u}$ with the non-diagonal matrix $T = \begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix}$. Derive the density $p_x(\mathbf{x})$ in terms of p_u, stating $\det T$, T^{-1}, and the final formula.

Question 31: Autoregressive Sequence Probability

An autoregressive model $p_w(\mathbf{x})$ models sequences (x_1, x_2, x_3) where each element is A, B, or C:

$$p_w(x_1) = \tfrac{1}{3}, \qquad p_w(x_2 \,|\, x_1) = \begin{cases} \frac{1}{2} & x_2 = x_1 \\ \frac{1}{4} & x_2 \neq x_1 \end{cases}, \qquad p_w(x_3 \,|\, x_2, x_1) = \begin{cases} \frac{1}{2} & x_3 = x_2 \\ \frac{1}{4} & x_3 \neq x_2 \end{cases}.$$

Compute $p_w(\mathbf{x})$ for $\mathbf{x} = (A, A, C)$, rounded to three decimals.

Answer: ________________

Bayesian Deep Learning

Question 32: Deep Ensembles and Marginalization

The main difference between the frequentist and Bayesian approach is the marginalization over parameters when defining the probability of a class label given input $\mathbf{x}$ and dataset $\mathcal{D}$:

$$p(\mathbf{y}^{\text{test}} \mid \mathbf{x}^{\text{test}}, \mathcal{D}) = \mathbb{E}_{\mathbf{w} \sim p(\mathbf{w} \mid \mathcal{D})}\left[p(\mathbf{y}^{\text{test}} \mid \mathbf{w}, \mathbf{x}^{\text{test}})\right].$$

Deep Ensembles approximate this full expectation with (weighted) averages. Which of the following are valid approximations? (Select one or more.)

- ☐ $\frac{1}{M}\sum_{m=1}^{M} p(\mathbf{y}^{\text{test}} \mid \mathbf{w}_m, \mathbf{x}^{\text{test}})$, where $\mathbf{w}_m$ are the weights of the m-th network of the ensemble.
- ☐ $p(\mathbf{y}^{\text{test}} \mid \mathbf{w}, \mathbf{x}^{\text{test}})$ for a single trained network.
- ☐ $\frac{1}{M}\sum_{m=1}^{M} \alpha_m \, p(\mathbf{y}^{\text{test}} \mid \mathbf{w}_m, \mathbf{x}^{\text{test}})$ with $\alpha_m = \text{softmax}(\mathbf{z})_m$ for some random vector $\mathbf{z}$.
- ☐ $\frac{1}{M}\sum_{m=1}^{M} p(\mathbf{y}_m^{\text{test}} \mid \mathbf{w}, \mathbf{x}_m^{\text{test}})$, where m denotes the m-th equally-sized fraction of the test data.

Graph Neural Networks

Question 33: Distinguishing Two Small Graphs

Consider two graphs: $\mathcal{G}_1$ is a path of three nodes (a coloured centre node joined to two leaves), and $\mathcal{G}_2$ is a star with a centre node joined to three leaves. Mark the correct statements. (Select one or more.)

- ☐ $\mathcal{G}_1$ and $\mathcal{G}_2$ can be distinguished by the Weisfeiler–Lehman (colour refinement) test.
- ☐ All message-passing neural networks (MPNNs) can distinguish $\mathcal{G}_1$ and $\mathcal{G}_2$.
- ☐ $\mathcal{G}_1$ and $\mathcal{G}_2$ are isomorphic graphs.

Question 34: Weisfeiler–Lehman Necessary Condition

Two graphs A and B are given. Do they fulfil the necessary condition for graph isomorphism given by the Weisfeiler–Lehman test (i.e. do they produce the same stable colour histogram)?

- ☐ True
- ☐ False

Question 35: Matching Graphs to Adjacency Matrices

Four graphs on six nodes are shown: A is a plain 6-cycle (hexagon), B is two triangles, C is another 6-cycle drawn differently, and D is a 6-cycle with one extra chord. For each adjacency matrix, name the matching graph (A, B, C, D, or none).

- All six row-sums equal 2 and the edges form a single closed cycle. __________
- Two row-sums equal 1 and the rest equal 2 (an open path, not a cycle). __________
- All six row-sums equal 2 but the edges split into two separate 3-cycles. __________
- Four row-sums equal 2 and two equal 3 (a cycle plus one chord). __________

Question 36: Group Action in Graph NNs

Which of the following is a common group action used in graph neural networks?

- ☐ Rotation
- ☐ Scaling
- ☐ Translation (permutation / relabelling of nodes)
- ☐ Cropping
- ☐ Clustering

Question 37: Equivariance of an Attention-Pooling Layer

A layer implements $\mathbf{h}^{(l+1)} = M \underbrace{\mathrm{softmax}(\beta\, M^{\mathsf{T}} \mathbf{h}^{(l)})}_{=:\mathbf{p}}$, where $\mathbf{h}^{(l)} \in \mathbb{R}^d$, $\beta > 0$ is a hyperparameter, and $M \in \mathbb{R}^{d \times J}$ holds J pattern columns. Which statements about this layer are true? (Select one or more.)

- ☐ $\mathbf{h}^{(l+1)}$ is equivariant to permutations of the columns of M.
- ☐ $\mathbf{p}$ is invariant to permutations of the columns of M.
- ☐ $\mathbf{p}$ is equivariant to permutations of the columns of M.
- ☐ $\mathbf{h}^{(l+1)}$ is invariant to permutations of the columns of M.
- ☐ $\mathbf{p}$ is equivariant to permutations of the rows of M.
- ☐ This layer cannot be used in deep learning because the gradient $\partial \mathbf{h}^{(l+1)} / \partial \mathbf{h}^{(l)}$ cannot be backpropagated.

Theory & Foundations

Question 38: Universal Approximator Theorem

Mark the correct statements about the Universal Approximator Theorem (UAT) for neural networks. (Select one or more.)

- ☐ For a real-valued, continuous function h on $[-\pi, \pi]^2$, the existence of an approximating network can be derived with the help of Fourier series.
- ☐ The UAT is a theoretical statement about the expressive power of neural networks.
- ☐ The UAT gives a useful algorithm to compute the optimal weights of a network that should approximate a given continuous function.
- ☐ The algorithm related to the UAT is easy to understand and implement, but very costly and therefore hardly used in practice.
- ☐ The UAT shows that any function can be approximated by any given network with arbitrary precision; there are no restrictions on the function or on the network structure.

Question 39: Newton's Method vs. Gradient Direction

On a quadratic error surface (elliptical level curves), two arrows are drawn from a point: one points straight at the minimum at the centre of the ellipses, the other points perpendicular to the local level curve. Label each arrow as *Gradient* or *Newton's Method*.

- Arrow pointing directly to the centre (minimum)

  ____

- Arrow perpendicular to the level curve (steepest descent)

  ____

Answer Key

Q1. **c, e.** AlexNet *does* have fully-connected layers (a wrong); it uses both strided convolutions *and* max-pooling (b wrong); it uses ReLU, not BatchNorm, which came later (d wrong).

Q2. **a, d.** VGG uses stacked 3×3 convolutions (two 3×3 replace one 5×5 at the same receptive field, so b is wrong); it is a plain classifier, not an encoder–decoder (c wrong); most parameters sit in the large FC layers.

Q3. **c, d.** AlexNet used Local Response Normalization, not BatchNorm (a wrong); LayerNorm is mostly used in transformers / sequence models, while CNN classifiers still use BatchNorm (b wrong).

Q4. **a.** Normalization is central to training deep ResNets (b wrong); pre-trained ResNets are widely used for transfer learning (c wrong); residual blocks also work with FC layers (d wrong); leaky ReLUs can be used (e wrong).

Q5. **b.** A 1×1 conv with 16 kernels keeps the spatial size and sets the channel depth to 16: output $[128, 128, 16]$. Stride 2 or 2×2 pooling would shrink the spatial resolution.

Q6. Increase stride: **no increase** (only the last layer changes, so its stride does not enlarge each neuron's RF). Dilated convolution: **strong increase**. Number of kernels: **no increase**. Kernel size: **small increase**.

Q7. **c** (0.5). With all weights/biases zero, every pre-activation is 0 and $\sigma(0) = 0.5$ at each layer.

Q8. **b** (may be positive or negative). The hidden activations are all the identical positive value 0.5, so the rows of $\mathbf{W}^{(2)}$ receive equal updates, but their sign is set by the per-output error $(\hat{y} - y)$, which can be positive or negative. (They are not all zero, so d is wrong.)

Q9. **a.** Pixel-wise labels are needed. This is semantic segmentation, not object detection (b wrong); CNNs with a (cross-entropy) loss are appropriate, not GANs (c, d wrong); IoU/mIoU is a standard segmentation metric (e wrong).

Q10. $\text{IoU} = \dfrac{|\text{intersection}|}{|\text{union}|} = \dfrac{5}{10 + 8 - 5} = \dfrac{5}{13} \approx \mathbf{0.385}$.

Q11. **a, b, d, e.** Faster R-CNN replaces selective search with an RPN; RoI-pooling/align act on feature maps; detection is a multi-task regression + classification problem; Fast R-CNN has a regression head and a classification head. (c is debatable/considered false: R-CNN runs a CNN on each warped proposal but is not characterised by "AlexNet" in the lecture.)

Q12. Faster R-CNN **two**-stage; YOLO **one**-stage; Mask R-CNN **two**-stage; Fast R-CNN **two**-stage; R-CNN **two**-stage; SSD **one**-stage.

Q13. $\mathbf{h} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 4 & 6 \\ 4 & 12 & 9 \end{pmatrix}$ (full stride-1 transpose convolution of two 2×2 inputs gives a 3×3 output).

Q14. $\mathbf{c} = \begin{pmatrix} 0 & 1 & 0 & -2 \\ 2 & 3 & -4 & -6 \\ 0 & -1 & 0 & 2 \\ -2 & -3 & 4 & 6 \end{pmatrix}$ (stride-2 transpose convolution spaces the kernel copies two apart, giving a 4×4 output).

Q15. Top row = GAN: trapezoid in = **Discriminator**, trapezoid out = **Generator**. Middle row = (V)AE: down = **Encoder** $q(\mathbf{z} \mid \mathbf{x})$, up = **Decoder** $p(\mathbf{x} \mid \mathbf{z})$. Bottom row = flow: first rectangle = **Flow**, second = **Inverse Flow**.

Q16. **e.** Only "encoder + decoder are parametrized networks" always holds. The lower-triangular Jacobian (a) is not generally true; mapping to a *distribution* (b) is the VAE; AEs need not be fully-connected (c) nor undercomplete (d) — sparse / overcomplete AEs exist.

Q17. **a, b, c.** Sparsity is enforced by a regularization term that keeps most *hidden* units inactive, and such an AE can still be trained on a copy task. It does not silence *input* units (d wrong), does not rely on early stopping (e wrong), and sparsity and denoising can be combined (f wrong).

Q18. 6. $u = \mathrm{ReLU}([\,1-2,\,1\,]) = \mathrm{ReLU}([-1,1]) = [0,1]$; $\hat{\mathbf{x}} = \mathrm{ReLU}([0,0,0]) = [0,0,0]$; loss $= (1{-}0)^2 + (1{-}0)^2 + (2{-}0)^2 = 1 + 1 + 4 = 6$. (If a $\frac{1}{n}$ MSE is used instead of the sum, the value is 2.0.)

Q19. **c.** VAEs maximize the ELBO, a lower bound on the data log-likelihood. Encoder/decoder need not be symmetric (a); the prior is typically a continuous (multivariate Normal) distribution, not discrete (b); they do not give $p(\mathbf{x})$ cheaply (d); they can generate images (e).

Q20. **d.** The ELBO is $\mathbb{E}_{q_\phi(\mathbf{z}\,|\,\mathbf{x})} \log p_\theta(\mathbf{x}\,|\,\mathbf{z}) - D_{\mathrm{KL}}(q_\phi(\mathbf{z}\,|\,\mathbf{x}) \,\|\, p_\theta(\mathbf{z}))$. Option b has the wrong KL arguments.

Q21. **a.** The KL-optimal Gaussian matches the first two moments: $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x})$ and $\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{x})$. Option e uses the second moment $\mathbb{E}(\mathbf{x}\mathbf{x}^\mathsf{T})$, which overestimates the covariance (it omits $-\mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x})^\mathsf{T}$).

Q22. **d.** The standard GAN value function is $\mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(1 - D(G(\mathbf{z})))]$.

Q23. **b, c.** The generator learns the data distribution and can produce unseen samples. The discriminator only learns real-vs-fake, not apple-vs-non-apple (a wrong); the discriminator loss reaching a constant is not guaranteed in general (d wrong).

Q24. **a, c, d, e.** A divergence must be non-negative and zero iff the distributions are equal, but *need not be symmetric* — so b (non-negative + symmetric) is the definition of a distance, not a divergence.

Q25. KL-divergence: grows quadratically without bound (**purple, solid**). Wasserstein-2: grows *linearly* (**green, long-dashed**). Jensen–Shannon: *saturates* at a finite value (**orange, short-dashed**).

Q26. **a, b, e.** FID compares the mean and covariance of real vs. generated features (so it detects mode collapse via the covariance) and is in fact the closed-form Wasserstein-2 distance between two Gaussians; IS and FID assess GAN image quality. They use Inception features, not AlexNet (c wrong), and are not training objectives (d wrong).

Q27. **d.** A lower-triangular Jacobian makes the determinant the product of the diagonal, which is cheap. Standard MLPs are generally not invertible, but it is not "impossible" to design invertible ones (a wrong); not all equal-dimension MLPs are flows (b); lower-triangular maps can be invertible (c).

Q28. **b, c.** Flows are fit by maximum likelihood, i.e. minimizing a KL-divergence; the general Jacobi determinant costs $\mathcal{O}(D^3)$. A uniform base *is* sufficient, so a is wrong; an autoregressive flow's Jacobian is triangular but its determinant is *not* always 1 (d wrong).

Q29. **b.** $p_x(\mathbf{x}) = \dfrac{1}{|\det T|}\, p_u(T^{-1}\mathbf{x})$ with $T^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}$ and $\det T = 2 \cdot \frac{1}{2} = 1$, so the prefactor is 1.

Q30. $\det T = 2\cdot 1 - (-2)\cdot 0 = 2$, so $T^{-1} = \frac{1}{2}\begin{pmatrix}1 & 2\\ 0 & 2\end{pmatrix} = \begin{pmatrix}\frac{1}{2} & 1\\ 0 & 1\end{pmatrix}$, giving $p_x(\mathbf{x}) = \dfrac{1}{|\det T|}\, p_u(T^{-1}\mathbf{x}) = \frac{1}{2}\, p_u\left(\begin{pmatrix}\frac{1}{2} & 1\\ 0 & 1\end{pmatrix}\mathbf{x}\right)$.

Q31. $p_w(A, A, C) = \frac{1}{3}\cdot\frac{1}{2}\cdot\frac{1}{4} = \frac{1}{24} \approx \mathbf{0.042}$ ($x_2 = A = x_1 \Rightarrow \frac{1}{2}$; $x_3 = C \neq A = x_2 \Rightarrow \frac{1}{4}$).

Q32. **a, c.** A uniform average over the ensemble members (a) and a weighted average (c) both approximate the posterior expectation. A single network (b) is not a Bayesian average; splitting the *test data* into fractions (d) is not ensembling.

Q33. **a.** They have different structure (in fact different node counts / degree sequences), so the WL test distinguishes them. Not all MPNNs can (expressivity depends on the aggregation), so b is wrong; they are not isomorphic (c wrong).

Q34. **True.** The two graphs produce the same stable WL colour histogram, so the necessary condition for isomorphism is fulfilled.

Q35. Single 6-cycle (all degrees 2, one cycle) $\to A$ (and the differently-drawn cycle $\to C$); open path (two degree-1 nodes) $\to$ *none of the above*; two separate 3-cycles $\to B$; cycle with one chord (two degree-3 nodes) $\to D$. Match each matrix by its row-sums (node degrees) and whether the edges form one cycle, two triangles, or a cycle-plus-chord.

Q36. **c** (Translation / permutation). GNNs are permutation invariant/equivariant; rotation, scaling, cropping, and clustering are not the defining group action here.

Q37. **c, d.** Permuting the columns of M permutes the rows of M^{T} and hence the entries of $\mathbf{p}$ (so $\mathbf{p}$ is *equivariant* to column permutations, c). Then $\mathbf{h}^{(l+1)} = M\mathbf{p}$ is unchanged because the column reorder of M cancels the row reorder of $\mathbf{p}$ (so $\mathbf{h}^{(l+1)}$ is *invariant*, d). The layer is differentiable (f wrong).

Q38. **a, b.** The UAT can be argued via Fourier series and is a statement about expressive power. It does *not* give an algorithm for the weights (c, d wrong), and it does have conditions (the function is continuous on a compact set and the network needs at least one hidden layer), so e is wrong.

Q39. Arrow to the centre = **Newton's Method** (uses curvature to point straight at the minimum); arrow perpendicular to the level curve = **Gradient** (steepest-descent direction).